

Cybersecurity Challenges in the Electric Vehicle Market

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SUMMARY

The electric vehicle (EV) market represents a small but rapidly growing market within the larger automotive industry. The market share of EVs has steadily increased in the past decade. It's also important to note that the growth in EVs applies not only to North America but also to other regions including Europe, Asia and Africa. The growing demand for electric vehicles has been significant because of many factors. A few of the key factors that had contributed to the significant growth in EV adoption are favourable government policies such as tax credits, increased competition, changing consumer dynamics, increased need for efficiency in transportation, and the need for environmentally friendly solutions. It's important to note that, with the growing number of EVs, their manufacturers have experienced many cyberattacks. The reported security incidents in the EV market imply that the EV industry is not prepared to address emerging cyberthreats. The emerging cyberthreats in the EV market are related to vulnerabilities in the network communications, the increased use of the cloud by the EV makers, the high reliance on third-party suppliers, the recent increase in ransomware attacks, the vulnerabilities involved in the internet of things and, most important, the charging methods and locations, which are becoming an increased concern. The recent data breaches further highlight the need to address the potential attack surface of an EV and the cybersecurity challenges in the EV market.

The growth in cyberattacks in the EV market raises questions of whether the EV manufacturers are addressing the cybersecurity challenges at sufficient capacity by adopting best practices. An inability of EV makers to adopt best practices in cybersecurity could lead to bigger impacts, potentially to the extent of affecting national security. The current security compliance standard that governs the automotive industry is ISO/SAE 21434:2021. Also known as "Road Vehicles – Cybersecurity Engineering," the standard focuses on establishing cyber resilience through seven different areas related to motor vehicle manufacturing. Those areas are cybersecurity management, concept development phase, product development phase, production, operations, maintenance, and end of cybersecurity support and decommissioning. It is important to understand the cybersecurity challenges and best practices in cybersecurity to help EV manufacturers better prepare for cybersecurity challenges. The increased focus on building cyber resilience can not only help EV makers attain sustainable growth but also ensure any potential threat to national security is minimized.

KEYWORDS

Cybersecurity
EV
Electric Vehicle
Charging Stations
EV Charging Stations
Power Grid
Open Charge Point Protocol
ISO/SAE 21434:2021
Security Architecture

INTRODUCTION

The sales volumes for electric vehicles have spiked, and the electric vehicle market has been growing drastically since 2011. In contrast with traditional vehicles, which work by burning gasoline or diesel fuel, EVs are powered by electricity stored in a rechargeable battery¹. The growing sales for EVs has motivated many large vehicle manufacturers to expand their product portfolios to include EVs². Over the past decade, many large vehicle manufacturers, such as VW Group, Toyota and Ford, have entered the electric vehicle market, increasing the competition among the EV players. The diversity of the EV product portfolio has increased from cars to SUVs and trucks. The growth in EV sales is applicable not only to North America but also to Africa, Asia, Eastern Europe, Latin America and Caribbean markets³. The published research indicates that the locations that subsidize the electric vehicle purchase most are the states of New York and California and the countries Spain and France. As supply chain risks are an important aspect to consider among the prevalent cybersecurity challenges in the EV market, the data indicates that China dominates the downstream of the EV battery supply chain. Moreover, the attack surface of an EV is large, while there are many other cybersecurity challenges to take into consideration. The cybersecurity challenges in the EV space expand from supply chain risks, charging infrastructure-related threats and insider threats. The published data suggests that the EV industry is at a growth stage in the industry life cycle, inducing more competition and innovation in the market. The barriers to entry seem low due to large government incentive schemes, encouraging more and more automotive giants to enter the EV space.

A CLOSER LOOK AT THE EV ARCHITECTURE, CHARGING INFRASTRUCTURE & ATTACK SURFACE

The EV architecture consists of many areas, including the internet of things and devices connected to the network. While the cyberthreats related to the network are quite common, despite the nature of the industry, it's impossible to ignore the role played by an EV's unique charging infrastructure. The nature of the charging infrastructure is one of the most discussed topics in cybersecurity. The charging infrastructure has aligned with a common standard called the open charge point interface (OCPI) protocol⁴. The U.S Department of Energy lists the following three areas related to the charging infrastructure:

¹ "The Road to an Electric Vehicle Future." Energy.Gov, 23 June 2023, <https://www.energy.gov/articles/road-electric-vehicle-future>.

² U.S. Bureau of Labor Statistics, February 2023.

³ Global Electric Vehicle Outlook, <https://www.iea.org/reports/global-ev-outlook-2022>

⁴ "Developing Infrastructure to Charge Electric Vehicles." Alternative Fuels Data Center, 2023, https://afdc.energy.gov/fuels/electricity_infrastructure.html.

1. Station location: A station location is a site with one or more EVSE (electric vehicle supply equipment) ports at the same address. Examples include a parking garage or a mall parking lot.
2. EVSE port: An EVSE port provides power to charge only one vehicle at a time, even though it may have multiple connectors. The unit that houses EVSE ports is sometimes called a charging post, which can have one or more EVSE ports.
3. Connector: A connector is what is plugged into a vehicle to charge it. Multiple connectors and connector types (such as CHAdeMO and CCS) can be available on one EVSE port, but only one vehicle will charge at a time. The connectors are sometimes called plugs.

The following diagram helps illustrate how the charging stations and the charging infrastructure work in an EV.

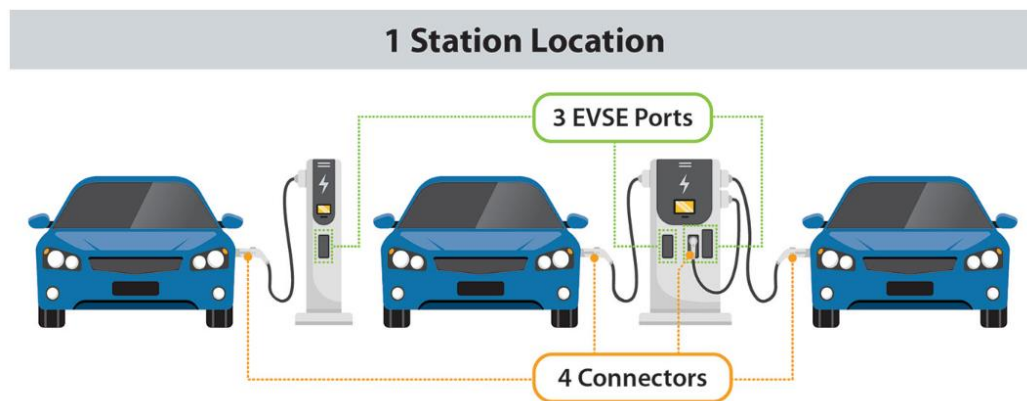


Figure 1: EV charging infrastructure. Source: U.S. Department of Energy

The diagram illustrates the nature of cybersecurity challenges related to the charging infrastructure, as there are many possibilities for a bad actor (threat actor)⁵ to tamper with an EV through the network and the connectors. It is reasonable to state that the charging infrastructure of an EV emphasizes the importance of the protection of automotive electronic systems, communication networks, control algorithms, software, users, and underlying data from malicious attacks, damage, unauthorized access, or manipulation.

When looking at the charging infrastructure and the nature of the communications, it's questionable if secure protocols are being used. Therefore, it's essential to note that there are several ways to architect EV charge points to ensure cybersecurity challenges are minimized. For instance, from using simple connectivity protocols such as built-in cellular modems, to using dedicated industrial gateways serving multiple charge points, steps can be taken to ensure the charging infrastructure is more secure from cyberattacks. Additionally, collecting large numbers of EV charge points and routing them through a single location increases the risk of a single point of failure.

When focusing on the attack surface of a charging infrastructure, it's important to note the vulnerabilities related to the digital systems and networks. The connected networks, such as mobile apps and cloud-based platforms, increase the exposure of an EV to a cyberattack. The possible attacks here can vary from unauthorized access and data breaches to the introduction of malware or ransomware. The intentions of a bad actor can vary from financial incentives to gaining control over critical functions, compromising the vehicle's safety mechanisms, or tampering with sensitive data. Another key feature of an EV is over-the-air updates. The over-the-air updates are crucial for

⁵ Editor, CSRC Content. "Threat Actor - Glossary: CSRC." CSRC Content Editor, 2023, https://csrc.nist.gov/glossary/term/threat_actor.

enhancing performance and adding new features to EVs, but if it is not sufficiently secured, it can serve as an attack vector for the bad actors. The nature of a cyberattack through over-the-air updates can include unauthorized modifications, system disruptions, and the installation of malicious software to tamper with an EV. It's important to note that such tampering can also result in events such as the theft of credit card information, resulting in an adverse financial impact on the end user.

In examining the charger-to-charger network, as discussed previously, the chargers are required to establish communication with a charging network using the open charge point protocol (OCPP), which ensures a secure connection. The few of the key characteristics that ensure security in the charging network are the capability to receive and implement secure software updates remotely while performing real-time protocol translation; secure encryption and decryption of the data; as well as authentication and authorization of the users during their communication with charging networks. As the charging networks are also responsible for conducting remote charger monitoring, diagnostics, control, and smart charge management, it's important to ensure that secure measurement, communication, storage, and reporting are in place. The secure and timely reporting of the energy and power, real-time charging-port status, real-time customer pricing, and historical charging-port uptime can minimize cybersecurity challenges and protect data from tampering.

While there are many aspects of cybersecurity risks to look at in the charging infrastructure and charging stations, some of the key actions that can ensure cybersecurity challenges are minimized are the implementation of a network protocol version management or extensive additional private network mitigations to reduce the possibility of reconnaissance or hacking. This is because many charging networks are discoverable using open-source tools. This discoverability implies a need for improving the back-end cloud as well as the front-end infrastructure. The presence of such security controls can minimize reconnaissance or hacking.

At a simpler level, the charging stations can be secured by ensuring there are physical barriers to entry against any intruders. The locked cabinets and rooms to deny physical access, the closure of access ports that are not required for operation, and reporting tampering or attempted vandalism of the equipment or tamper-indicating seals can act as proactive measures in preventing a cyberattack. Furthermore, at a broader level, regular security audits, collaboration with cybersecurity experts, and prompt patching of vulnerabilities are crucial in ensuring the security and integrity of electric vehicles. The compromise of a charging station can result in adverse events such as the theft of personal information, including payment data, location, or demographic information. This can happen through cyberattacks such as, denial-of-service attacks, SQL injection attacks, and cross-site scripting (XSS) attacks in which the bad actor exploits vulnerabilities within the charging infrastructure.

Even in the absence of reported incidents of an EV being compromised, there remains the possibility of a larger event occurring in future as the EV market continues to grow dramatically. It's important to note that the automobile industry is electrifying, with an expected market capitalization of \$457 billion in 2023, and the EV market is expected to nearly double by 2027 with expected vehicle sales of 16 million per year⁶.

Cybersecurity Challenges Related to the Power Grid

When examining the power grid, one of the main risks is the possibility of instability due to an external event caused by a bad actor. There is a possibility that a bad actor could tamper with the EV charging infrastructure through its connection to the power grid. The tampering could happen through any of the ways mentioned in the previous section, such as a denial-of-service attack. If a bad actor compromises the power grid through the charging infrastructure connection during peak times, it could lead to voltage fluctuations, transformer overload, and potential power outages or other disruptions to

⁶ Dorot, Uri. "EV Charging Station Applications – a Growing Cyber Security Risk." Radware Blog, 1 May 2023, https://www.radware.com/blog/application-protection/2023/05/ev_charging_station_cyber_threats/.

the electrical grid. Another possibility is that a bad actor could exploit vulnerabilities within the charging infrastructure to overload the grid and manipulate power flows. This illuminates how critical it is for energy providers to ensure proper load management and coordination between charging stations while maintaining security controls. Failure could also result in a national security threat and disrupt the economy. For example, Ukrainian hackers remotely disrupted and shut down EV stations in Russia while displaying anti-Putin messages in 2022⁷.

Therefore, to properly protect EV charging applications, infrastructure, and their connections to the electric grid, it's important to have effective collaboration among infrastructure providers, automakers, utility providers, and government entities. It's also critical to ensure necessary upgrades are being made, EV makers are adopting security frameworks such as NIST, ISO/SAE 21434:2021, and implementing innovative solutions that can help manage EV charging infrastructure and the power grid effectively.

THE NATURE OF DATA BREACHES IN THE EV MARKET

Cybersecurity challenges in the electric vehicle industry are on the rise despite the rising demand and increased investment in the EV space. Most data breaches are a result of a compromise of an internal system of an EV manufacturer, rather than an attack on a single EV. The potential compromise of a billion-dollar EV company can offer a higher financial incentive for a bad actor than a compromise of a single EV. Competition is a common contributor to the data breaches that have occurred in the electric vehicle industry. This means there are situations in which an existing employee of an EV manufacturer sells company secrets to the competition. Data breaches in the electric vehicles are also a result of the lack of internal security controls. Many of these security controls are related to the lack of identity access management. The following sections include detailed analysis of recent data breaches in the EV market.

Data Loss Prevention, Background Checks, and the Implementation of Least Privilege Principle

On Feb. 3, 2021, Tesla reported an internal data breach in which sensitive data was leaked by a newly hired software engineer named Alex Khatilov. Khatilov had access to more information than he needed to. He transferred the company-sensitive data into his Dropbox, and this was not detected by Tesla. Khatilov conducted this illegitimate activity within two weeks of joining the company. The security incident emphasizes the need to implement strict data loss prevention practices to detect the transfer of sensitive data. It raises questions of whether Tesla had stringent data loss prevention practices in place. Furthermore, with the growth in insider threats, it's important to redesign how employee background checks are conducted. In the wake of such attacks, the traditional method of screening only the employee's criminal background may not be sufficient. For instance, using only the Social Security number for background checks cannot detect a potential employee's activity in the dark web. There are more advanced tools, such as cyber threat intelligence, which are able to scan for an individual's activity in the dark web. This can help an organization stay secure and minimize potential insider threats. This security incident also emphasizes the importance of the least privilege principle. As NIST defines the least privilege principle, "Security architecture should be designed so that each entity is granted the minimum system resources and authorizations that the entity needs to perform its function."⁸ The Khatilov security incident implies a violation of the least privilege principle.

⁷ Lambert, Fred. "Hacked Electric Car Charging Stations in Russia Display 'Putin Is a D*ckhead' and 'Glory to Ukraine.'" Electrek, 1 March 2022, <https://electrek.co/2022/02/28/hacked-electric-car-charging-stations-russia-displays-putin-dckhead-glory-to-ukraine/>.

⁸ CSRC - Computer Security Resource Center. "Least Privilege - Glossary: CSRC." CSRC NIST, 2023, https://csrc.nist.gov/glossary/term/least_privilege.

Local Admin Privileges and the Need to Scan for 'Ghost Accounts' Even After an Employee Leaves the Organization

Many recent data breaches suggest that Tesla has become a lucrative target for the bad actors. On June 22, 2018, a security incident was reported in which a former employee named Martin Tripp leaked sensitive information, consisting of a video and images, about Tesla's manufacturing systems to a third party. It's unknown if the information was transferred to a competitor. Tripp used a hacking software which helped in exfiltrating information, and the software was tracking internal information even after Tripp had left the company. There were reports that Tripp was a disgruntled employee who was upset about not receiving a promotion. This security incident highlights a few important factors. One aspect to consider here is why and how endpoint detection systems failed to capture the data exfiltration. The employee had continued access to the information even after leaving the organization, which reveals Tesla was not scanning for activity generated from dormant accounts. Another aspect to consider is how Tripp had local admin rights to install the software to exfiltrate gigabytes of data. Local admin rights pose a high risk to an organization, especially with employees who deal with sensitive data daily. The local admin rights should have been disabled or gone through an approval process before being granted to the employee. Also, the monitoring of such rights could have prevented the data breach.

Improved Vendor Risk Assessments that Are Business Case-Specific Rather than Process-Specific

It's important to conduct due diligence before onboarding a new vendor. This is also a security compliance control in many security industry standards. It was reported on March 13, 2021, that Tesla had experienced a data breach through a third-party vendor called Verkada. Verkada is based in California and develops cloud-based building security and operating systems. Verkada combines security equipment (such as video cameras, access control systems, and environmental sensors) with cloud-based machine vision and artificial intelligence. The breach occurred through a compromise of a super admin user account on the vendor's end. The hackers were able to get access into the super admin user account and then view the security cameras of many companies, such as Tesla and Cloudflare. This raises questions about why a vendor was given super admin rights to a customer's security cameras and if that was sufficiently addressed in the vendor's contract. It's important to evaluate the business case against the security standards to identify any loopholes, and then address it through contractual terms in the agreement. It is also important to examine a vendor's breach notification policy and the privacy policy. This identifies how long a vendor will take to notify a customer in an event of a data breach and what information the vendor has access to. Typically, attackers conduct reconnaissance before conducting an attack. Enforcement of a breach notification policy could have helped Tesla be notified of any minor security incidents that could have led to a larger attack later.

Growing Ransomware Attacks in the EV Industry

While there is a growing trend of cyberattacks involving insider threats and the theft of sensitive data to sell to the competition, it's worthwhile to note that in the recent past, the EV industry has also been hit by ransomware attacks. On Jan. 4, 2023, it was reported that Volvo was hit by a ransomware attack. In addition, on Dec. 21, 2022, it was reported that Nio was hit by a ransomware attack. Insufficient details on these attacks are publicly available to identify how the systems were compromised. However, as the EV industry grows, it's important to note that the hackers are more motivated by the financial incentives from a ransomware attack. For instance, it was published that the attack on Nio involved a ransom demand of \$2.25 million. This also highlights the importance of enforcing regulations in the market to ensure customer data is protected from cyberattacks.

MINIMIZING CYBERSECURITY CHALLENGES IN THE EV INDUSTRY THROUGH COMPLIANCE AND REGULATORY STANDARDS

The data breaches and the susceptibility of an EV architecture to a cyberattack discussed in the previous sections suggest a powerful need to adopt best practices in cybersecurity. The regulations and standards for the EV market, such as ISO/SAE 21434, provide a security framework that EV makers can adopt. The standard outlines the cyber risk management under seven different stages involved in vehicle manufacturing.

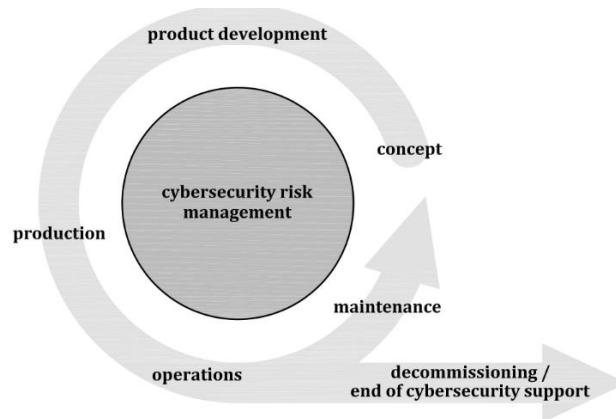


Figure 2: Overall cybersecurity risk management. Source: ISO/SAE 21434

The adoption of ISO/SAE 21434 can help protect internal systems and customer data from cyberattacks. However, it's important to highlight that there is no enforcement of a security standard for the EV market at this time, and it's almost impossible to identify which EV makers are certified under ISO/SAE 21434. This can also be considered a gap in the current EV market.

The best practices an EV manufacturer can adopt to minimize a cyberattack include implementing vulnerability management, best practices in data protection, data security controls, timely software updates, best practices for network communications, physical security, utilizing threat intelligence to better prepare, and complying with existing regulatory requirements. Vulnerability management can be implemented efficiently by EV makers by establishing a process to identify and mitigate vulnerabilities in a timely manner. This includes conducting regular security assessments and implementing security patches as soon as they are available. An EV maker can also implement best practices in data protection. The protection of data is important, as it can involve items such as customer data collected from the vehicle's sensors and cameras. EV makers need to implement strong encryption protocols to ensure that data such as customer information, vehicle telemetry data, and software updates are protected. Strong access controls to prevent unauthorized access to data becomes important as well. The implementation of necessary software updates can help address vulnerabilities and security issues in a timely manner. In addition, network communication security becomes an important factor in ensuring the data is transmitted in a secure manner. The EV makers need to ensure that communication between the vehicle and other devices, such as charging stations and traffic lights, is secure. This includes using strong encryption, strong transport layer level security, and authentication mechanisms to prevent unauthorized access to these channels.

Finally, supply chain risks can be minimized through the adoption of a compliance framework such as ISO/SAE 21434. As discussed in the charging infrastructure section, physical security is an important element in ensuring customer data is protected. For instance, the implementation of physical security controls, such as access control and video surveillance, can minimize the risk of cyberattacks that happen through the compromise of physical infrastructure. The threat intelligence plays a vital role in implementing proactive security controls against bad actors. As cyberattacks are becoming more

sophisticated, the EV makers need to collect and analyse threat intelligence in a timely manner to identify and respond to potential cyberattacks.

The implementation of regulatory and compliance standards — such as National Highway Traffic Safety Administration (NHTSA) and the Federal Trade Commission (FTC), UN Regulation 155, NHTSA Cybersecurity Guidance, NIST Cybersecurity Framework, the General Data Protection Regulation (GDPR), and the California Consumer Privacy Act (CCPA) — can also provide a minimum required layer of security against bad actors.

CONCLUSION

The EV is known as the “vehicle of the future.” There is increased demand for EVs, growing market share for EVs worldwide, and large tax benefits granted for EV purchases driving up the demand even more. However, the cybersecurity challenges in the EV space are growing because of the nature of the architecture of an EV and a lack of adoption of best security practices by the EV makers. There is also a potential national security threat as EV market share keeps increasing, and therefore it’s important to address the cybersecurity challenges in a timely manner. There is a long road for EV makers in addressing those challenges due to the lack of enforcement of a security framework on them. The nature of cyberattacks today is based on increased competition, however this could change in the future as the EV use keeps growing rapidly. The adoption of best practices such as ISO/SAE 21434 is critical for security, and the time has come for EV makers to take cybersecurity challenges seriously.

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